

Lecture 10

The BFV Scheme

SEC 595: Encrypted Computing

Roadmap

Lecture 8 defined homomorphic encryption and Lecture 7 built an LWE scheme that only encrypts. This lecture assembles the first lattice scheme that adds and multiplies: BFV (Fan and Vercauteren, 2012). Every ingredient already exists — the ring R_q (Lecture 3), the scaling factor Δ (Lecture 1), gadget decomposition (Lecture 4), and RLWE (Lecture 6). Sections 1–4 give the spaces, encoding, keys, encryption and the correctness condition. Section 5 shows that addition is cheap. Sections 6–7 treat multiplication, its noise growth, and relinearization. Section 8 runs everything on a small instance. Sections 1–5 form the first session and Sections 6–8 the second.

1. Spaces and Parameters

1.1 The Interface

Primitive	Returns
ParamGen(λ)	(n, t, q) and the distributions
KeyGen	SK (kept), PK and EK (published)
Encrypt(PK, M)	$\text{ct} \in R_q \times R_q$
Decrypt(SK, ct)	$M \in R_t$
EvalAdd($\text{ct}^{(1)}, \text{ct}^{(2)}$)	an encryption of $M^{(1)} + M^{(2)}$
EvalMult($\text{ct}^{(1)}, \text{ct}^{(2)}$)	an encryption of $M^{(1)}M^{(2)}$
Relinearize(EK, ct^*)	a two-element ct encrypting the same M

BFV is **levelled** (Lecture 8): it evaluates circuits up to a multiplicative depth fixed when (n, t, q) is chosen. Relinearize computes nothing; it repairs the shape of the output of EvalMult.

1.2 Two Rings

Definition 1 (Plaintext and ciphertext spaces). Let n be a power of two and $q \gg t \geq 2$. Plaintexts lie in $R_t = \mathbb{Z}_t[X]/(X^n + 1)$ and ciphertexts in $R_q \times R_q$, where $R_q = \mathbb{Z}_q[X]/(X^n + 1)$.

Coefficients are taken in the centered range, and $[\cdot]_q$ denotes reduction into it (Lecture 1). For $v \in R_q$, $\|v\|$ is the largest absolute coefficient, and $\|ab\| \leq n\|a\|\|b\|$ (Lecture 3); every noise bound below is an application of this inequality. Since $|R_t| = t^n$ and $|R_q \times R_q| = q^{2n}$, at $n = 4$, $t = 4$, $q = 2^{14}$ there are $4^4 = 256$ plaintexts against 2^{112} ciphertexts: encryption has ample room to be randomised.

Example 1 (Plaintexts at $(n, t) = (4, 5)$). For $t = 5$ the centered range is $\{-2, -1, 0, 1, 2\}$, and a

plaintext is a polynomial of degree at most 3 with coefficients in it. Hence

$$1 + 2X + X^2 - X^3, \quad -1 - 2X - X^2 + 2X^3, \quad 1 + X + X^2 + X^3$$

all lie in R_5 , and $|R_5| = 5^4 = 625$.

1.3 Distributions and Parameter Choice

Symbol	Distribution	Used for
R_2	ternary, coefficients uniform in $\{-1, 0, 1\}$	SK and u
$\mathcal{X}$	discrete Gaussian (μ, σ, β)	e, e_1, e_2
R_q	uniform over R_q	the mask a

The standard choice is $\mu = 0$, $\sigma = 8/\sqrt{2\pi} \approx 3.2$ and tail-cut $\beta = \lceil 6\sigma \rceil = 19$ (Lecture 6, §1.3). Because of the tail-cut, $\|e\| \leq \beta$ is treated as a hard bound, and noise analysis becomes arithmetic.

The security parameter λ is a work parameter: $\lambda = 128$ means the best known attack costs about 2^{128} operations. The rule of thumb is to minimise n , t and q subject to the computation fitting. A larger n costs time and size and buys security; a larger q buys depth but costs security at fixed n ; a larger t buys plaintext room but shrinks the budget $q/2t$ and inflates the growth factor $2tn^2$ of Section 6.

2. Encoding

Encoding maps data into R_t . It uses no keys and no randomness.

Naive encoding puts m in the constant term, $M = m + 0 \cdot X + \dots + 0 \cdot X^{n-1}$, and decodes by reading it back. It wastes $n - 1$ coefficients, and the single coefficient must hold every intermediate value, so t must exceed the largest one — and a large t costs noise budget.

Definition 2 (Base- k encoding). Write $m = (a_{n-1} \dots a_1 a_0)_k$ and set $M = a_{n-1}X^{n-1} + \dots + a_1X + a_0$. Decode by evaluating, $m = M(k)$, reading coefficients as non-negative representatives in $\{0, \dots, t - 1\}$.

Coefficients now start at most $k - 1$ instead of m . But multiplication raises degrees, and degree wraps at n .

Proposition 1 (Compatibility). *Base- k encoding is compatible with a circuit when, at every intermediate result, (a) no coefficient reaches t in absolute value, and (b) the degree stays below n .*

If (a) fails, a coefficient is corrupted modulo t . If (b) fails, $X^n \equiv -1$ folds high terms onto low ones with a sign flip. Boolean circuits over $\mathbb{Z}_2$ tolerate (a) by design, since reduction modulo 2 is the intended semantics.

Example 2 (Encoding by hand, $n = 4$). *Base 2.* $13 = 1101_2$, so $M = 1 + X^2 + X^3$, and $M(2) = 1 + 4 + 8 = 13$.

Base 3. $13 = 111_3$, so $M = 1 + X + X^2$, and $M(3) = 1 + 3 + 9 = 13$.

Base 2 has coefficients at most 1, so it leaves more room before (a) fails; base 3 uses degree 2 instead of 3, so it leaves more room before (b) fails.

$m = 21$ in base 2. $21 = 10101_2$ needs X^4 . In R_t , $1 + X^2 + X^4 = 1 + X^2 - 1 = X^2$, which decodes to 4, not 21.

3. Key Generation

Definition 3 (KeyGen). Sample $SK \leftarrow R_2$, $a \leftarrow R_q$ and $e \leftarrow \mathcal{X}$, and set

$$PK_0 = [- (a \cdot SK + e)]_q, \quad PK_1 = a.$$

The key identity is one line:

$$PK_0 + PK_1 \cdot SK = -(aSK + e) + aSK = -e. \quad (1)$$

The public key, evaluated at the secret, is small. The pair (PK_0, PK_1) is one RLWE sample, so distinguishing it from uniform is decision RLWE (Lecture 6, §2.2). The evaluation key EK is also produced here; it is defined in Section 7.

4. Encryption and Decryption

4.1 The Scaling Factor

Let $\Delta = \lfloor q/t \rfloor$ and write $q = \Delta t + [q]_t$ with $0 \leq [q]_t < t$. Multiplying by Δ places the message in the top $\log_2 t$ bits and leaves the bottom $\log_2(q/t)$ bits for noise (Lecture 1). Decryption scales by t/q and *rounds*. The residue $[q]_t$ vanishes exactly when $t \mid q$.

4.2 The Algorithms

Definition 4 (Encrypt and Decrypt). $\text{Encrypt}(PK, M)$: sample $u \leftarrow R_2$ and $e_1, e_2 \leftarrow \mathcal{X}$, and return $ct = (c_0, c_1)$ with

$$c_0 = [PK_0 \cdot u + e_1 + \Delta M]_q, \quad c_1 = [PK_1 \cdot u + e_2]_q. \quad (2)$$

$\text{Decrypt}(SK, ct)$: return

$$M = \left[\left[\frac{t [c_0 + c_1 \cdot SK]_q}{q} \right] \right]_t. \quad (3)$$

The three small polynomials u, e_1, e_2 are drawn per encryption and never needed again. They make encryption randomised, as IND-CPA requires (Lecture 8, §1.5).

4.3 The Decryption Identity

Proposition 2 (Decryption identity). *For a fresh ciphertext,*

$$c_0 + c_1 \cdot SK = \Delta M + v \pmod{q}, \quad v = -e \cdot u + e_1 + e_2 \cdot SK.$$

Proof. Substitute (2) and then the definition of PK:

$$\begin{aligned} c_0 + c_1 SK &= -(aSK + e)u + e_1 + \Delta M + (au + e_2)SK \\ &= -auSK - eu + e_1 + \Delta M + auSK + e_2SK = \Delta M - eu + e_1 + e_2SK. \quad \square \end{aligned}$$

The mask $au\text{SK}$ cancels. The encryptor creates it inside PK_0u without knowing SK ; the decryptor creates it inside $c_1\text{SK}$ without knowing u . What remains is the scaled message plus the **noise** v .

Proposition 3 (Fresh noise). *If $\|e\|, \|e_1\|, \|e_2\| \leq \beta$ and $\|u\|, \|\text{SK}\| \leq 1$, then $\|v\| \leq n\beta + \beta + n\beta = 2n\beta + \beta$. If u and SK are only β -bounded, $\|v\| \leq 2n\beta^2 + \beta$.*

Proof. Apply $\|ab\| \leq n\|a\|\|b\|$ to eu and $e_2\text{SK}$, and add $\|e_1\| \leq \beta$. □

At $n = 4096$, $\beta = 19$:

$$2n\beta + \beta = 155\,667 \approx 2^{17.25}, \quad 2n\beta^2 + \beta = 2\,957\,331 \approx 2^{21.50}.$$

The second, conservative bound is the one quoted in the literature, because it also covers variants with $u \leftarrow \mathcal{X}$. At $\beta = 1$ the two coincide.

4.4 The Correctness Condition

Proposition 4 (Correctness). *If $c_0 + c_1\text{SK} = \Delta M + v \pmod{q}$ with $M \in R_t$ and*

$$\|v\| < \frac{q}{2t},$$

then Decrypt returns M .

Proof. Over $\mathbb{Z}$, $c_0 + c_1\text{SK} = \Delta M + v + qr$ for some integer polynomial r . Scaling by t/q gives $\frac{t\Delta}{q}M + \frac{t}{q}v + tr$, and $\frac{t\Delta}{q} = 1 - \frac{[q]t}{q}$ differs from 1 by a negligible amount. Rounding removes $\frac{t}{q}v$ when each coefficient is below $\frac{1}{2}$, which is the hypothesis; reduction modulo t removes tr . □

Only the rounding can fail, and the failure is *silent*: a coefficient comes back wrong, with no warning. The gap between $\|v\|$ and $q/2t$ is the **noise budget**; every homomorphic operation spends part of it.

Example 3 (One failure). Take $t = 8$, $q = 1024$, one coefficient $m = 5$. Then $\Delta = \lfloor 1024/8 \rfloor = 128$, $\Delta m = 640$, and the bound is $q/(2t) = 1024/16 = 64$.

$$\begin{aligned} v = 30 : \quad & \frac{8}{1024}(640 + 30) = \frac{5360}{1024} = 5.234375 \longrightarrow 5 \quad \checkmark \\ v = 70 : \quad & \frac{8}{1024}(640 + 70) = \frac{5680}{1024} = 5.546875 \longrightarrow 6 \quad \times \end{aligned}$$

At $v = 70 > 64$ the coefficient comes back as 6. The output is a valid element of $\mathbb{Z}_8$, so the decryptor cannot tell.

5. EvalAdd

Definition 5 (EvalAdd).

$$\text{EvalAdd}(\text{ct}^{(1)}, \text{ct}^{(2)}) = \left([c_0^{(1)} + c_0^{(2)}]_q, [c_1^{(1)} + c_1^{(2)}]_q \right).$$

The result has the shape of a fresh encryption of $M^{(1)} + M^{(2)}$ with randomness $u^{(1)} + u^{(2)}$ and $e_i^{(1)} + e_i^{(2)}$, so it can be fed back in.

Proposition 5 (Additive noise growth). $c_0^{(3)} + c_1^{(3)}\text{SK} = \Delta(M^{(1)} + M^{(2)}) + (v_1 + v_2)$, so $\|v_{\text{add}}\| \leq \|v_1\| + \|v_2\|$.

Proof. Add the two decryption identities. $\square$

Thus L fresh ciphertexts sum to noise at most $L(2n\beta + \beta)$. At $n = 4096$, $\beta = 19$, fresh noise is about $2^{17.25}$, so a budget of 2^{178} permits about 2^{160} additions. To add a plaintext M' , add $\Delta M'$ to c_0 and leave c_1 ; the noise is unchanged. Addition is essentially free: the whole difficulty is multiplication.

6. EvalMult

6.1 Multiplying the Identities

Read ct as $c_0 + c_1 S \in R_q[S]$, so that decryption evaluates at $S = \text{SK}$ (Lecture 4). Over $\mathbb{Z}$, $\text{ct}^{(i)}(\text{SK}) = \Delta M^{(i)} + v_i + q r_i$. Multiplying,

$$\begin{aligned} (\text{ct}^{(1)}\text{ct}^{(2)})(\text{SK}) &= \Delta^2 M^{(1)} M^{(2)} + \Delta(M^{(1)} v_2 + M^{(2)} v_1) + q(v_1 r_2 + v_2 r_1) \\ &\quad + q\Delta(M^{(1)} r_2 + M^{(2)} r_1) + v_1 v_2 + q^2 r_1 r_2. \end{aligned} \tag{4}$$

The leading term carries Δ^2 , so one factor must come off.

6.2 Scaling by t/q

Dividing by Δ fails on $q^2 r_1 r_2$: $\Delta \nmid q^2$ in general, and the non-integral residue survives reduction modulo q . Scaling by t/q instead sends $q^2 r_1 r_2 \mapsto q t r_1 r_2$, which vanishes modulo q . Since $t/q \approx 1/\Delta$, the leading term still comes out right. Because t/q is not an integer, the result must be rounded, adding an error r_e .

Write $M^{(1)} M^{(2)} = [M^{(1)} M^{(2)}]_t + t r_M$ and $v_1 v_2 = [v_1 v_2]_\Delta + \Delta r_v$. Substituting into (4) and scaling by t/q gives, modulo q ,

$$\begin{aligned} \frac{t}{q}(\text{ct}^{(1)}\text{ct}^{(2)})(\text{SK}) &= \Delta[M^{(1)} M^{(2)}]_t + (M^{(1)} v_2 + M^{(2)} v_1) + t(v_1 r_2 + v_2 r_1) + r_v \\ &\quad - [q]_t(r_M + M^{(1)} r_2 + M^{(2)} r_1) + r_e. \end{aligned}$$

This has the standard shape $\Delta M^{(3)} + v_3$ with $M^{(3)} = [M^{(1)} M^{(2)}]_t$. The dominant term of v_3 is $M^{(1)} v_2 + M^{(2)} v_1$: the messages multiply the noises.

Proposition 6 (Multiplicative noise growth). *Up to lower-order terms,*

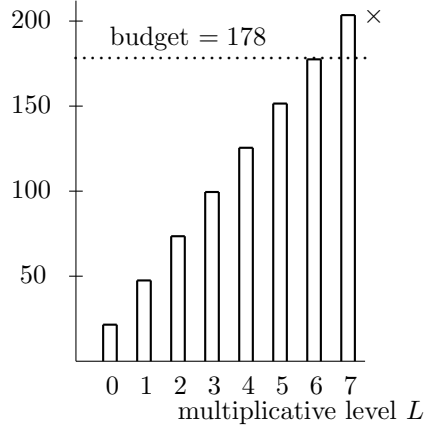
$$\|v_{\text{mult}}\| \lesssim \|v_i\| \cdot G, \quad G = 2 t n^2 \|\text{SK}\|,$$

where $\|v_i\|$ is the larger input noise.

Proof sketch. Since c_0 and c_1 have centered coefficients below $q/2$, $c_0 + c_1 \text{SK}$ has coefficients of size at most about $nq\|\text{SK}\|$, so $\|r_i\| \lesssim n\|\text{SK}\|$. The term $t(v_1 r_2 + v_2 r_1)$ then has norm at most $t \cdot 2 \cdot n \|v_i\| \cdot n \|\text{SK}\| = \|v_i\| G$. The other terms are smaller. $\square$

After L sequential multiplications the noise is about $\|v_0\| G^L$: exponential in the **depth**, and in nothing else. At $n = 4096$, $t = 2$ and ternary SK, $G = 2 \cdot 2 \cdot 4096^2 = 67\,108\,864 = 2^{26}$, i.e. 26 bits of budget per level.

6.3 A Depth Budget



Bar heights are $\log_2\|v\| = 21.50 + 26L$ after L multiplications at $n = 4096$, $t = 2$, $\beta = 19$, $q = 2^{180}$. The dotted line is the budget $\log_2(q/2t) = 178$, cleared by half a bit at $L = 6$ and overshoot by 25.5 bits at $L = 7$.

Example 4 (How many multiplications?). Take $n = 4096$, $t = 2$, $\beta = 19$, ternary SK, $q = 2^{180}$, and the conservative fresh-noise bound.

(a) *Noise and growth.* $\|v_0\| \leq 2n\beta^2 + \beta = 2 \cdot 4096 \cdot 361 + 19 = 2957331 \approx 2^{21.50}$, and $G = 2 \cdot 2 \cdot 4096^2 \cdot 1 = 2^{26}$.

(b) *Depth.* The budget is $q/(2t) = 2^{180}/4 = 2^{178}$. We need $21.50 + 26L < 178$, i.e. $L < 156.50/26 \approx 6.02$, so $L = 6$. Check: $21.50 + 6 \cdot 26 = 177.50 < 178$ ✓, while $21.50 + 7 \cdot 26 = 203.50 > 178$ ×.

(c) *Depth 12.* We need $21.50 + 12 \cdot 26 = 333.50 < \log_2 q - 2$, so $\log_2 q > 335.50$, i.e. $\log_2 q \geq 336$.

(d) *Larger t.* At $t = 2^{16}$, $G = 2 \cdot 2^{16} \cdot 2^{24} = 2^{41}$ and the budget is $2^{180}/2^{17} = 2^{163}$. Then $21.50 + 41L < 163$ gives $L < 3.45$, so $L = 3$.

6.4 The Procedure

Definition 6 (EvalMult). Tensor, scale by t/q , round and reduce:

$$\text{ct}^* = \left(\left[\left[\frac{t c_0^{(1)} c_0^{(2)}}{q} \right] \right]_q, \left[\left[\frac{t(c_0^{(1)} c_1^{(2)} + c_1^{(1)} c_0^{(2)})}{q} \right] \right]_q, \left[\left[\frac{t c_1^{(1)} c_1^{(2)}}{q} \right] \right]_q \right).$$

The three components are the coefficients of S^0, S^1, S^2 in $(c_0^{(1)} + c_1^{(1)} S)(c_0^{(2)} + c_1^{(2)} S)$. They are formed over $\mathbb{Z}$ from centered representatives *before* scaling; reducing modulo q first would destroy the information the division by q needs.

The output has two problems: it has **three** ring elements instead of two, and it decrypts only through $c_0^* + c_1^* \text{SK} + c_2^* \text{SK}^2$, which needs SK^2 . Both are repaired by relinearization.

Multiplying by a plaintext M' is cheap: $(M'c_0, M'c_1)$ needs no tensoring and no rescaling, and the noise grows to at most $n\|M'\|\|v\|$, not by the factor G .

7. Relinearization

7.1 The Evaluation Key

Given $\text{ct}^* = (c_0^*, c_1^*, c_2^*)$ we want $\hat{\text{ct}} = (\hat{c}_0, \hat{c}_1)$ with

$$c_0^* + c_1^* \text{SK} + c_2^* \text{SK}^2 \approx \hat{c}_0 + \hat{c}_1 \text{SK} \pmod{q}.$$

The evaluator cannot compute $c_2^* \text{SK}^2$, so SK^2 is supplied in masked form.

Definition 7 (Evaluation key). Sample fresh $a \leftarrow R_q$ and $e \leftarrow \mathcal{X}$, and publish

$$\text{EK} = (\text{EK}_0, \text{EK}_1) = \left([- (a \cdot \text{SK} + e) + \text{SK}^2]_q, a \right).$$

As in (1),

$$\text{EK}_0 + \text{EK}_1 \cdot \text{SK} = \text{SK}^2 - e. \quad (5)$$

EK is an encryption of SK^2 under SK, so BFV needs a circular-security assumption on top of RLWE.

7.2 Basic Relinearization

Definition 8 (Relinearize, basic form). $\hat{c}_0 = [c_0^* + \text{EK}_0 \cdot c_2^*]_q$ and $\hat{c}_1 = [c_1^* + \text{EK}_1 \cdot c_2^*]_q$.

Proposition 7. $\hat{c}_0 + \hat{c}_1 \text{SK} = c_0^* + c_1^* \text{SK} + c_2^* \text{SK}^2 - c_2^* \cdot e$.

Proof. $\hat{c}_0 + \hat{c}_1 \text{SK} = c_0^* + c_1^* \text{SK} + c_2^* (\text{EK}_0 + \text{EK}_1 \text{SK})$; apply (5). $\square$

The error $-c_2^* e$ is a disaster. The key error e is small, but c_2^* is a ciphertext component spread over all of $\mathbb{Z}_q$, so

$$\|c_2^* \cdot e\| \leq n \|c_2^*\| \beta \approx \frac{1}{2} n q \beta,$$

which exceeds q for $n \geq 2$. Basic relinearization destroys the ciphertext.

7.3 Gadget Decomposition

Fix a base w and let $\ell = \lceil \log_w q \rceil$. Decompose c_2^* coefficientwise (Lecture 4):

$$c_2^* = \sum_{i=0}^{\ell-1} w^i d_i, \quad \|d_i\| < w.$$

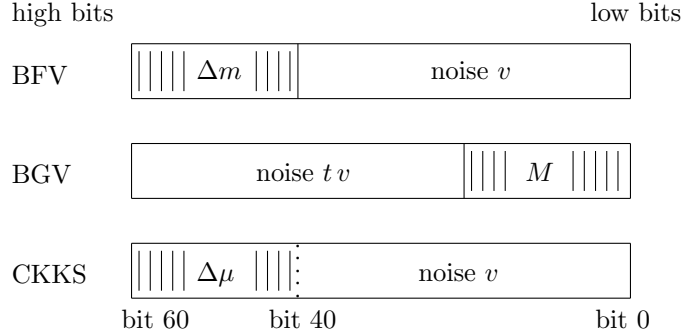
Definition 9 (Relinearize with ℓ keys). Publish $\text{EK}^{(i)} = ([-(a_i \text{SK} + e_i) + w^i \text{SK}^2]_q, a_i)$ for $0 \leq i < \ell$, and set $\hat{c}_j = [c_j^* + \sum_i \text{EK}_j^{(i)} d_i]_q$ for $j = 0, 1$.

Proposition 8. $\hat{c}_0 + \hat{c}_1 \text{SK} = c_0^* + c_1^* \text{SK} + c_2^* \text{SK}^2 - \sum_i d_i e_i$, and $\|\sum_i d_i e_i\| \leq n \ell w \beta$.

Proof. $\text{EK}_0^{(i)} + \text{EK}_1^{(i)} \text{SK} = w^i \text{SK}^2 - e_i$, so the sum contributes $\sum_i w^i d_i \text{SK}^2 - \sum_i d_i e_i = c_2^* \text{SK}^2 - \sum_i d_i e_i$. Each $\|d_i e_i\| \leq n w \beta$, and there are ℓ terms. $\square$

A blow-up of order q is traded for an *additive* term of order w , at ℓ times the key material and ℓ times the work. Since EvalMult has just multiplied the noise by G , the relinearization error is almost never the binding constraint. Choosing w trades key size against noise. Relinearization is an instance of **key switching**, here from SK^2 to SK ; the same machinery rotates slots in BGV and CKKS.

8. Where the Message Sits



The word is $\log_2 q = 60$ bits wide with $t = 2^{20}$, $\Delta = 2^{40}$; hatching marks the message. BFV puts it in the top twenty bits, BGV in the bottom twenty, CKKS in the top with a dotted boundary, there being no separation at all.

	BFV	BGV	CKKS
Decryption	$c_0 + c_1 SK$ $= \Delta m + v$	$c_0 + c_1 SK$ $= M + tv$	$c_0 + c_1 SK$ $= \Delta\mu + v$
Recover by	scaling by t/q , rounding, then mod t	reducing mod t ; no rounding	dividing by Δ and accepting the error
Correct when	$\ v\ < q/(2t)$	$\ v\ < q_\ell/(2t)$	never exactly; precision about $\log_2 \Delta - \log_2 \ v\ $, in $\mathbb{C}^{n/2}$

5.1 Reading the Table

BFV keeps q fixed and tracks the invariant noise tv/q , scaling down by t/q after each multiplication; at $q = 2^{30}$, $t = 2^{10}$ the message holds the top ten bits under $\Delta = 2^{20}$ and the budget is 2^{19} . BGV's error always appears as $t \cdot e$, so it never touches the message's residue modulo t and decryption is a reduction, at the cost of noise growing toward $q_\ell/2$ and shrunk by modulus switching from q_ℓ to $q_{\ell-1} = q_\ell/p_\ell$: at $q_\ell = 2^{60}$, $t = 2^{20}$ and $\|v\| = 2^{35}$ against a budget 2^{39} , switching by $p_\ell = 2^{20}$ leaves 2^{15} against 2^{19} , the same four-bit gap, and each $p_i \equiv 1 \pmod{t}$ so the message survives. CKKS does not separate message from noise: the plaintext is $\Delta\mu$ under the canonical embedding of Lecture 4 and v costs $\log_2 \|v\|$ bits of precision, so $\Delta = 2^{40}$ with $\|v\| = 2^{12}$ keeps 28 bits, and RESCALE by $p_\ell = 2^{40}$ undoes a multiplication's scale squaring and divides the noise in one step — so its decryption returns μ plus the error, carrying information about SK .

9. A Worked Instance

8.1 Setup and Encryption

Example 5 (The instance). Take $n = 4$, $t = 4$, $q = 2^{14} = 16384$, $\beta = 1$, and

$$\text{SK} = 1 - X + X^3, \quad e = 1 - X^2 + X^3, \quad a = 7561 + 3002X + 15115X^2 + 8443X^3.$$

Parameters. $\Delta = 16384/4 = 4096$ and $t \mid q$, so $[q]_t = 0$. The budget is $q/(2t) = 16384/8 = 2048$. Fresh noise is at most $2n\beta + \beta = 9$, and $G = 2 \cdot 4 \cdot 16 \cdot 1 = 128$. One multiplication is guaranteed: $9 \cdot 128 = 1152 < 2048$, and even with the relinearization bound 112 of Section 8.3, $1152 + 112 = 1264 < 2048$. A second is not: $1152 \cdot 128 = 147456 > 2048$. Only one level is guaranteed.

Public key. $\text{PK}_0 = [-(a\text{SK} + e)]_q = 3381 + 3290X + 12715X^2 + 15494X^3$ and $\text{PK}_1 = a$; one checks $\text{PK}_0 + \text{PK}_1\text{SK} = -1 + X^2 - X^3 = -e$.

Encode. In base 2, $m_1 = 5$ gives $M^{(1)} = 1 + X^2$ and $m_2 = 2$ gives $M^{(2)} = X$.

Encrypt. With $u^{(1)} = 1 - X^2 + X^3$, $e_1^{(1)} = X - X^2$, $e_2^{(1)} = 1 - X$, and $u^{(2)} = -1 + X + X^2$, $e_1^{(2)} = 1 - X^3$, $e_2^{(2)} = -X + X^2 + X^3$, (2) gives (coefficients in $[0, q)$)

$c_0^{(1)}$	$518 + 6070X + 14319X^2 + 15585X^3$
$c_1^{(1)}$	$3291 + 12713X + 15495X^2 + 13002X^3$
$c_0^{(2)}$	$1179 + 5077X + 10340X^2 + 510X^3$
$c_1^{(2)}$	$1649 + 12499X + 11833X^2 + 9675X^3$

Decrypt. Computing $[c_0 + c_1\text{SK}]_q - \Delta M$, and independently $-eu + e_1 + e_2\text{SK}$, gives the same noise in each case:

$$v_1 = 2 - 3X + 3X^2 - X^3, \quad \|v_1\| = 3, \quad v_2 = 4 - 2X - X^2 + X^3, \quad \|v_2\| = 4.$$

Both are below 9, so (3) returns $M^{(1)}$ and $M^{(2)}$, which decode to 5 and 2.

8.2 One Addition, One Multiplication

Example 6 (Add and multiply). *EvalAdd.* The noise is $v_1 + v_2 = 6 - 5X + 2X^2$, with norm $6 \leq \|v_1\| + \|v_2\| = 7$. Decryption returns $1 + X + X^2$, which decodes to $1 + 2 + 4 = 7 = 5 + 2$.

EvalMult. The three components are

$$\begin{aligned} c_0^* &= 13721 + 1467X + 6265X^2 + 4702X^3, \\ c_1^* &= 11383 + 12209X + 15054X^2 + 14179X^3, \\ c_2^* &= 7501 + 6571X + 10312X^2 + 14554X^3. \end{aligned}$$

Evaluating $c_0^* + c_1^*\text{SK} + c_2^*\text{SK}^2$ gives $\Delta M^{(3)} + v_3$ with

$$M^{(3)} = X + X^3, \quad v_3 = 7 - 2X + X^2 + 2X^3, \quad \|v_3\| = 7.$$

$M^{(3)}$ decodes to $2 + 8 = 10 = 5 \times 2$. The noise grew from 4 to 7, against the worst case $\|v_2\| \cdot G = 4 \cdot 128 = 512$: worst-case bounds are very pessimistic, which is the price of a guarantee.

8.3 Relinearization

Example 7 (Relinearize, both ways). Take $a' = 11923 + 604X + 9981X^2 + 14002X^3$ and $e' = X - X^2$.

Key. $\text{EK}_0 = [-(a'\text{SK} + e') + \text{SK}^2]_q = 7450 + 4913X + 4626X^2 + 442X^3$, $\text{EK}_1 = a'$, and $\text{EK}_0 + \text{EK}_1\text{SK} - \text{SK}^2 = -X + X^2 = -e'$, as (5) states.

Basic. By Proposition 7 the new noise is $[v_3 - c_2^*e']_q = 4249 - 5673X + 931X^2 - 3739X^3$, so $\|v\| = 5673 > 2048$. Decryption returns 1 instead of $X + X^3$: the message is destroyed. The bound predicted it: in centered form $c_2^* = 7501 + 6571X - 6072X^2 - 1830X^3$, so $n\|c_2^*\|\beta = 4 \cdot 7501 = 30\,004$.

Gadget, $w = 4$. Here $\ell = \lceil \log_4 16384 \rceil = 7$, so seven keys. The base-4 digits of c_2^* (coefficients in $[0, q)$) are

$$\begin{aligned} d_0 &= 1 + 3X + 2X^3, & d_1 &= 3 + 2X + 2X^2 + 2X^3, & d_2 &= 2X + X^3, \\ d_3 &= 1 + 2X + X^2 + 3X^3, & d_4 &= 1 + X, & d_5 &= 3 + 2X + 2X^2 + 2X^3, \\ d_6 &= 1 + X + 2X^2 + 3X^3, \end{aligned}$$

and $\sum_i 4^i d_i = c_2^*$. For the toy, all seven keys reuse a' and e' (insecure, but the noise depends only on the e_i). Then $\sum_i d_i e_i = (\sum_i d_i) e'$ with $\sum_i d_i = 10 + 13X + 7X^2 + 13X^3$, so $\sum_i d_i e' = -6 + 23X + 3X^2 - 6X^3$ and the new noise is

$$v_3 - \sum_i d_i e' = 13 - 25X - 2X^2 + 8X^3, \quad \|v\| = 25,$$

against $\|v_3\| + n\ell w\beta = 7 + 4 \cdot 7 \cdot 4 = 119$. Decryption returns $X + X^3$. The relinearized ciphertext is

$$\hat{c}_0 = 8279 + 2761X + 4059X^2 + 9120X^3, \quad \hat{c}_1 = 1940 + 11017X + 7767X^2 + 811X^3.$$

8.4 Where It Breaks

Example 8 (A second multiplication). Multiply $\hat{\text{ct}}$ by $\text{ct}^{(2)}$ again, to compute $5 \times 2 \times 2 = 20$. Decrypting the three-element result gives noise $-8 - 5X - 19X^3$, of norm 19, far inside the budget 2048, and

$$M^{(4)} = [(X + X^3) \cdot X]_t = X^2 + X^4 = -1 + X^2 = 3 + X^2 \text{ in } R_4.$$

So $M^{(4)} = M^{(3)}M^{(2)}$ in R_t : BFV worked. But decoding at $X = 2$ gives $3 + 4 = 7$, not 20. The encoding failed: $20 = 10100_2$ needs degree 4, which $n = 4$ does not have (condition (b)), and the wrap $X^4 \equiv -1$ put $-1 \equiv 3 \pmod{4}$ in the constant term.

A computation can therefore fail in two independent ways — through the noise budget, or through the plaintext geometry — and neither announces itself. Both must be checked when the parameters are chosen.

Summary

1. Plaintexts lie in R_t and ciphertexts in $R_q \times R_q$; $\text{PK}_0 + \text{PK}_1\text{SK} = -e$. Base- k encoding needs coefficients below t and degree below n .
2. $c_0 + c_1\text{SK} = \Delta M + v$ with $v = -eu + e_1 + e_2\text{SK}$; fresh noise is at most $2n\beta + \beta$ (conservatively $2n\beta^2 + \beta$).
3. Decryption is correct while $\|v\| < q/2t$, and failure is silent.

4. EvalAdd adds noises. EvalMult scales the tensor product by t/q and multiplies the noise by $G = 2tn^2\|\text{SK}\|$, so depth is the scarce resource.
5. Relinearize absorbs $c_2^*\text{SK}^2$ using EK, at error $-c_2^*e$; gadget decomposition cuts it from about $nq\beta$ to $n\ell w\beta$.

Exercises

1. Compute $(1 + 2X + X^2 - X^3)(1 + X + X^2 + X^3)$ in R_5 with $n = 4$ (Example 1), using $X^4 \equiv -1$.
2. Encode $m = 6$ in base 2 at $n = 4$, square the polynomial in R_4 , and decode. Which compatibility condition fails, and how does the answer change at $n = 8$?
3. Verify (1) for the keys of Example 5. Then encrypt $M = 1 + X$ with $u = 1$, $e_1 = e_2 = 0$, decrypt, and compute v both from $[c_0 + c_1\text{SK}]_q - \Delta M$ and from $-eu + e_1 + e_2\text{SK}$.
4. Take $q = 100$, $t = 7$, so $\Delta = 14$ and $[q]_t = 2$. Show that $\Delta \nmid q^2$, and divide the term $q^2 r_1 r_2$ of (4) by Δ and by q/t in turn. Which result is removed by reduction modulo q ?
5. For $(n, t) = (4096, 2)$, $(4096, 256)$ and $(8192, 2)$ with ternary SK, compute G in bits and the depth supported at $q = 2^{180}$, $\beta = 19$, using the conservative fresh-noise bound. Which of n , t , q buys depth most cheaply?
6. In Example 5, how many fresh ciphertexts can be added before the worst-case bound $L(2n\beta + \beta)$ reaches $q/2t$? Compare with the guaranteed multiplicative depth.
7. For $q = 16384$, $n = 4$, $\beta = 1$, tabulate ℓ and $n\ell w\beta$ for $w = 2, 4, 16, 128$. What is paid in key material as w decreases? Then find the smallest power-of-two n and the smallest t for which $5 \times 2 \times 2$ decodes correctly under base-2 encoding.